

EMERGENCY ROOM PATIENT MANAGEMENT SYSTEM

Doubly Linked list Simulation

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Problem Statement

To simulate an Emergency Room Queue System using a Doubly Linked List, where:

- Critical patients are treated first (insert at beginning).
- Normal patients are added at the end.
- Special patients are admitted at a specific position.
- Treated patients are deleted from the beginning.

Proposed Solution

We use a Doubly Linked List (DLL) where each node represents a patient with:

- patient_id
- prev pointer (link to previous patient)
- next pointer (link to next patient)

Graphical Representation

